TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

Anonymous Author(s)
Affiliation
Address
email

Abstract

Operator learning methods typically approximate the solution map of a physical system with a single global model an assumption that fails when solutions change character fundamentally across the parameter space.

We propose TEMPO: a framework that identifies latent dynamical regimes in snapshot data via Expectation-Maximization, builds a dedicated basis for each, and learns to route new inputs to the right regime at inference time. On multi-regime benchmarks, TEMPO achieves substantially lower cross-regime errors than per-parameter specialists, with interpretable structure that reflects the underlying physics.

1 Introduction

Operator learning approximates the solution map of a parametric PDE from data, enabling instant prediction without re-solving the governing equations at each new parameter value [1]. The dominant paradigm exemplified by DeepONet [2] and POD-DeepONet [3] trains a single global model over the entire parameter space. This works well when solutions form a simple manifold, but breaks down when the governing parameter drives *qualitatively distinct* dynamics: viscosity separating laminar diffusion from near-discontinuous shocks, or source magnitude producing solutions that differ by orders of magnitude in amplitude. In these multi-regime settings, a single global basis must compromise between incompatible geometries, concentrating error precisely where accurate prediction matters most.

Several methods improve expressivity while retaining a single global representation. POD-PoU [4] trains a partition-of-unity mixture of POD bases with a shared branch, reducing error on sharp-gradient problems but without discovering the underlying regime structure. NeuralPOD [5] constructs nonlinear orthogonal basis functions via greedy residual minimization, representing each mode as a neural network and thus overcoming the linear subspace constraint. Both, however, operate on the full dataset globally and provide no operator for routing new inputs to the appropriate representation at inference.

Clustering snapshots and fitting local bases to each cluster is a classical idea in reduced-order modeling [6, 7]. Daniel et al. [8] extend this to deep learning by clustering solution subspaces on the Grassmann manifold and training a dictionary selector at inference. These methods use hard cluster assignments, operate offline on fixed datasets, and do not couple regime discovery with the operator learning objective leaving basis quality and assignment accuracy disconnected.

We propose **TEMPO** (*Trajectory EM Mixture of POD Operators*), which closes this gap in a principled probabilistic framework. TEMPO models the trajectory ensemble as a Gaussian mixture and recovers regime structure via Expectation-Maximization (EM): the E-step softly assigns each trajectory to regimes based on reconstruction quality, while the M-step rebuilds a dedicated NeuralPOD

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

basis per regime so as bases sharpen, assignments improve, and vice versa. After convergence, a GatingNet and per-regime BranchNets are trained jointly, routing new inputs to the correct regime at inference.

Contributions:

- A Gaussian mixture model with EM that discovers M latent regimes and fits a dedicated NeuralPOD basis per regime, requiring no regime labels.
- NeuralPOD-DeepONet: the first deployable operator built on the NeuralPOD basis of [5], pairing it with a branch network that predicts mode coefficients for unseen inputs.
- An adaptive basis update rule based on a distribution-shift criterion, which determines when each regime's basis warrants retraining and avoids redundant computation across EM iterations.
- On 1D Burgers' and 2D Darcy flow, a single TEMPO model trained jointly across all parameter values substantially outperforms single-regime baselines, with discovered regimes that reflect intrinsic solution geometry rather than parameter labels.

2 Problem Statement

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain and $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ a time interval.

We are given a dataset of N solution trajectories:

$$\mathcal{D} = \left\{ \left(u_0^{(i)},\, \kappa^{(i)},\, s^{(i)}\right) \right\}_{i=1}^{N}, \qquad s^{(i)} = s\Big(\cdot,\, \cdot;\, u_0^{(i)}, \kappa^{(i)}\Big) \in L^2(\Omega \times \mathcal{T}), \tag{1}$$

where $u_0^{(i)} \in \mathcal{U} = L^2(\Omega)$ is the initial condition and $\kappa^{(i)} \in \mathcal{K}$ the physical parameter. Each trajectory $s^{(i)}$ is discretized on $N_y = N_t N_x$ quadrature points $\{y_j = (x_j, t_j),\, w_j\}_{j=1}^{N_y}$ covering the product grid $\Omega \times \mathcal{T}$, with weighted norm $\|f\|_w^2 = \sum_j w_j f(y_j)^2$.

Definition 1 (Solution operator)**.** *The* solution operator

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \to L^2(\Omega \times \mathcal{T})$$

maps each pair (u_0, κ) *to the full space-time solution:* $\mathcal{G}(u_0, \kappa) = s(\,\cdot\,,\,\cdot\,;\, u_0, \kappa)$.

The operator $\mathcal{G}$ is unknown and expensive to evaluate directly. The *operator learning problem* is to construct $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ alone, generalizing to unseen $(u_0, \kappa) \notin \mathcal{D}$ without solving the PDE.

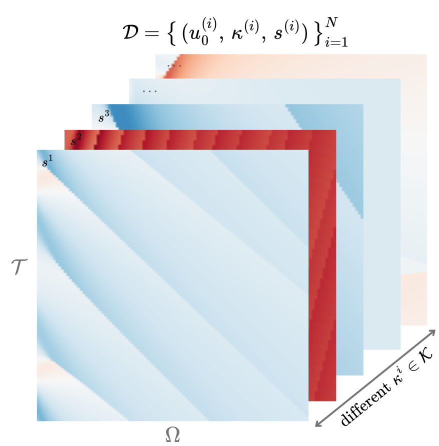

Figure 1: Training dataset $\mathcal{D}$: solution trajectories $s^{(i)}$ on $\Omega \times \mathcal{T}$ for different $\kappa^{(i)} \in \mathcal{K}$. Solutions change character fundamentally across parameter values.

Multi-regime assumption. When solutions change character fundamentally across $\mathcal{K}$, the optimal low-dimensional representation differs from regime to regime, and no single global basis captures the full solution diversity. We formalize this observation via a latent variable model over the trajectory ensemble.

Definition 2 (Generative model)**.** *Each trajectory arises from one of M latent regimes. The regime label $z_i \in \{1, \dots, M\}$ is unobserved and drawn from a categorical prior,*

$$z_i \sim \mathrm{Categorical}(\pi_1, \dots, \pi_M), \qquad \pi_m \geq 0, \quad \textstyle\sum_m \pi_m = 1. \tag{2}$$

Conditioned on $z_i = m$, the trajectory is distributed as

$$(s^{(i)} \mid z_i{=}m) \ \sim\ \mathcal{N}\big(\hat{s}_m^{(i)},\ \sigma^2 W^{-1}\big), \tag{3}$$

where $W = \mathrm{diag}(w_1, \dots, w_{N_y})$ is the quadrature weight matrix, $\sigma^2 > 0$ controls the softness of regime assignments, and $\hat{s}_m^{(i)}$ is the regime-m approximation to $s^{(i)}$.

The generative model leaves $\hat{s}_m^{(i)}$ unspecified. We require only that it admits a *low-rank trajectory decomposition*:

$$\hat{s}_m^{(i)}(y) \ =\ \phi_0^{(m)}(y) + \sum_{k=1}^{K_m} \lambda_k^{(m,i)}\, \phi_k^{(m)}(y), \qquad y = (x,t) \in \Omega \times \mathcal{T}, \tag{4}$$

where $\phi_k^{(m)} : \Omega \times \mathcal{T} \to \mathbb{R}$ are spatiotemporal mode functions and $\lambda_k^{(m,i)} \in \mathbb{R}$ a scalar amplitude for trajectory i on mode k. This form encompasses POD [3] and learned neural bases [5]; in this work we adopt the latter.

Definition 3 (Regime approximator)**.** *The parameters of regime m are $\Phi_m = \{\phi_k^{(m)}\}_{k=0}^{K_m}$ and scalar amplitudes $\Lambda_m = \{\lambda_k^{(m,i)}\}_{k=1,\, i=1}^{K_m,\, N}$, where $\phi_0^{(m)} : \Omega \times \mathcal{T} \to \mathbb{R}$ is the regime mean field, $\{\phi_k^{(m)}\}_{k\geq 1}$ spatiotemporal mode functions, $\lambda_k^{(m,i)} \in \mathbb{R}$ the scalar amplitude of trajectory i on mode k, and $K_m \leq K_{\max}$ the adaptive mode count.*

The learning problem decomposes into two phases: discovering the latent regime structure from $\mathcal{D}$ *offline*, then training a deployable operator for fast inference *online*.

Phase 1: Offline regime discovery. Since the regime labels z_i in (3) are unobserved, we recover the bases $\{\Phi_m\}$, mixture weights π, and noise level σ^2 by maximizing the observed-data log-likelihood:

$$(\Phi^*, \pi^*, \sigma^{2*}) \ =\ \operatorname*{arg\,max}_{\Phi,\, \pi,\, \sigma^2} \sum_{i=1}^{N} \log \sum_{m=1}^{M} \frac{\pi_m\, \det(W)^{1/2}}{(2\pi\sigma^2)^{N_y/2}} \exp\!\left(-\frac{\left\| s^{(i)} - \hat{s}_m^{(i)} \right\|_w^2}{2\sigma^2} \right). \tag{5}$$

The solution yields soft regime responsibilities $\gamma_{im}^* = p(z_i{=}m \mid s^{(i)}, \Phi^*, \pi^*, \sigma^{2*})$ that serve as supervision for Phase 2.

Phase 2: Online operator learning. With bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ frozen, we train a gating network (*GatingNet*) and M branch networks (*BranchNets*) that map a new input (u_0, κ) to a weighted combination of local regime predictions. The precise architecture and loss are given in Section 3.4. The quality of the final operator is measured by the mean relative L^2 error:

$$\varepsilon \ =\ \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\left\| s^{(i)} - \hat{s}(\cdot;\, u_0^{(i)}, \kappa^{(i)}) \right\|_w}{\left\| s^{(i)} \right\|_w}. \tag{6}$$

3 Method

All methods studied in this work share a common two-phase structure: an *offline* basis-fitting phase that constructs a low-rank approximation to the solution manifold from snapshots in $\mathcal{D}$, and an *online* operator-learning phase that trains a branch network to predict expansion coefficients from a new input (u_0, κ). The four methods differ along two independent axes:

- **Basis type.** A *linear* (POD) basis uses SVD of the (weighted) snapshot matrix; a *nonlinear* (NeuralPOD) basis learns spatiotemporal mode networks by sequential residual minimization.
- **Number of regimes.** A *single-regime* operator fits one global basis to all snapshots; a *multi-regime* operator (TEMPO) discovers M latent regimes via EM and assigns a dedicated basis to each.

This gives four combinations: POD-DeepONet [3] (linear, global), NeuralPOD-DeepONet (nonlinear, global, our extension of [5]), TEMPO(POD) (linear, M regimes), and TEMPO(NeuralPOD) (nonlinear, M regimes).

3.1 Basis Representations

3.1.1 POD Basis

Let $\phi_0 := \bar{s} = \frac{1}{N}\sum_i s^{(i)}$ be the mean field and $\tilde{S} \in \mathbb{R}^{N_y \times N}$ the matrix with columns $\tilde{s}^{(i)} = s^{(i)} - \phi_0$. The rank-$K$ truncated SVD $\tilde{S} \approx U_K \Sigma_K V_K^\top$ yields orthonormal modes $\{\phi_k\}_{k=1}^K$ (columns of U_K) and coefficients $\lambda_k^{(i)} = [\Sigma_K V_K^\top]_{ki}$; the approximation

$$\hat{s}^{(i)} = \phi_0 + \sum_{k=1}^{K} \lambda_k^{(i)}\, \phi_k \tag{7}$$

is optimal in the L^2 sense among all rank-K affine representations.

For TEMPO(POD), the M-step (17) is solved analytically: set $\phi_0^{(m)} := \bar{s}_m = \sum_i \gamma_{im} s^{(i)}$ and apply SVD to the weighted centered matrix with columns $\sqrt{\gamma_{im}}\,(s^{(i)} - \phi_0^{(m)})$. This is the closed-form linear analogue of Algorithm 1.

3.1.2 NeuralPOD Basis

To overcome the linearity constraint, we parametrize each mode function $\phi_k : \Omega \times \mathcal{T} \to \mathbb{R}$ as a Fourier feature network [9]; we call the resulting basis a *Fourier NeuralPOD* basis:

$$\phi_k(y) \;=\; f_{\theta_k}\big(\psi(y)\big), \qquad \psi(y) \;=\; \big[\sin(2\pi B y),\; \cos(2\pi B y)\big], \tag{8}$$

where $B \in \mathbb{R}^{F \times d}$ is a fixed random matrix drawn once at initialization and $f_{\theta_k} : \mathbb{R}^{2F} \to \mathbb{R}$ is a small MLP with Tanh activations. Plain MLPs exhibit spectral bias [10]; the Fourier encoding resolves this. For multi-scale coverage, the rows of B are partitioned evenly across L frequency scales $\sigma_1 < \cdots < \sigma_L$: the ℓ-th block is drawn from $\mathcal{N}(0,\, \sigma_\ell^2 I)$. The mean field ϕ_0 uses the same architecture.

The scalar amplitudes $\lambda_k^{(i)} \in \mathbb{R}$ are direct learnable parameters, one per training trajectory, optimized jointly with ϕ_k. After Phase 1 they are frozen and serve as regression targets for the BranchNet in Phase 2.

Modes are learned by *greedy residual minimization* [5]. Let $r^{(0,i)} = s^{(i)} - \phi_0$, where ϕ_0 is fitted to the snapshot mean $\bar{s}$ defined above. For each subsequent mode $k \geq 1$, given the residual

$$r^{(k-1,i)}(y) = s^{(i)}(y) - \phi_0(y) - \sum_{\ell=1}^{k-1} \lambda_\ell^{(i)}\, \phi_\ell(y), \tag{9}$$

we solve

$$\min_{\phi_k,\, \{\lambda_k^{(i)}\}} \sum_{i=1}^{N} \sum_j w_j \Big(\lambda_k^{(i)}\, \phi_k(y_j) - r_j^{(k-1,i)}\Big)^2 \tag{10}$$

(the global case with uniform weights; the γ-weighted generalisation used in TEMPO is Algorithm 1). Mode addition stops when the residual falls below fraction τ of the initial variance,

$$\sum_{i=1}^{N} \|r^{(k,i)}\|_w^2 \;<\; \tau \cdot \sum_{i=1}^{N} \|s^{(i)} - \bar{s}\|_w^2, \tag{11}$$

or $k = K_{\max}$. Each mode captures the largest remaining variance component the nonlinear analogue of Gram–Schmidt orthogonalization.

3.2 Single-Regime Operators

3.2.1 POD-DeepONet

POD-DeepONet [3] replaces the learned trunk network of DeepONet [2] with the fixed POD basis. After computing the global modes $\{\phi_k\}_{k=1}^K$ and training coefficients $\{\lambda^{(i)}\}$ offline, a branch network $\mathcal{B}_\theta : \mathbb{R}^{m_s+d_\kappa} \to \mathbb{R}^K$ is trained online by regression against these coefficients:

$$\mathcal{L}_{\text{branch}} = \frac{1}{N}\sum_{i=1}^{N}\big\|\mathcal{B}_\theta(u_0^{(i)},\, \kappa^{(i)}) - \lambda^{(i)}\big\|^2. \tag{12}$$

The prediction at any $(x,t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(x,t;\, u_0,\kappa) = \phi_0(x,t) + \sum_{k=1}^{K}\mathcal{B}_{\theta,k}(u_0,\kappa)\,\phi_k(x,t). \tag{13}$$

3.2.2 NeuralPOD-DeepONet

Mou et al. [5] introduce NeuralPOD as a dimensionality reduction for PDE solutions, establishing the greedy basis-fitting procedure of Section 3.1.2. However, they do not construct a deployable solution operator: no network is proposed to predict the mode coefficients for an unseen initial condition. We close this gap by substituting the NeuralPOD basis into the POD-DeepONet framework; we call this combination *NeuralPOD-DeepONet*.

The procedure follows POD-DeepONet exactly with the Fourier NeuralPOD basis substituted for POD. Offline fitting yields coefficients $\boldsymbol{\lambda}^{(i)} \in \mathbb{R}^K$; online, a branch network minimizes (12) and prediction follows (13). At equal mode count K, the nonlinear basis spans a strictly richer function space than POD.

3.3 TEMPO: Offline Regime Discovery

A single global basis whether linear or nonlinear must compromise between qualitatively distinct regimes, as the optimal basis geometry differs fundamentally between them. TEMPO avoids this by constructing a dedicated basis for each of M latent regimes, discovered from unlabelled trajectories via EM on the generative model of Definition 2.

Initialization. A global POD is computed on all N trajectories; each trajectory is projected to a P-dimensional coefficient vector $\alpha^{(i)} \in \mathbb{R}^P$. Running a standard GMM on $\{\alpha^{(i)}\}$ yields initial responsibilities $\gamma_{im}^{(0)}$, weights $\pi_m^{(0)}$, and per-regime statistics $(\mu_m^{(0)}, \Sigma_m^{(0)})$. Each regime basis $\Phi_m^{(0)}$ is then fitted to the responsibility-weighted ensemble: for TEMPO(NeuralPOD) via WEIGHTEDNEURALPOD (Algorithm 1); for TEMPO(POD) via the weighted SVD of Section 3.1. The noise level σ^2 is calibrated from the initial reconstructions rather than treated as a free hyperparameter:

$$\sigma^2 \;\leftarrow\; \frac{1}{2N}\sum_{i=1}^{N}\sum_{m=1}^{M}\gamma_{im}^{(0)}\,\big\|s^{(i)} - \hat{s}_m^{(i)}\big\|_w^2, \tag{14}$$

which ties the softness of the E-step directly to the quality of the initial bases.

E-step. At iteration q, the responsibility of trajectory i for regime m is updated by Bayes' rule on the Gaussian likelihood (3):

$$\gamma_{im}^{(q)} = \frac{\pi_m^{(q-1)}\,\exp\!\Big(-\dfrac{\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2}{2\sigma^2}\Big)}{\sum_{j=1}^{M}\pi_j^{(q-1)}\,\exp\!\Big(-\dfrac{\|s^{(i)} - \hat{s}_j^{(i)}\|_w^2}{2\sigma^2}\Big)}, \tag{15}$$

where $\hat{s}_m^{(i)}$ is evaluated via (4) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$. As the bases improve in the M-step, the reconstructions $\hat{s}_m^{(i)}$ sharpen and the E-step automatically refines regime assignments in response distinguishing TEMPO from a GMM applied to a fixed feature space. When κ drives large amplitude variation (as with $\beta \in [0.01, 100]$ in the Darcy experiments), replacing the squared norm with the relative squared norm $\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2 / \|s^{(i)}\|_w^2$ in both (15) and (14) gives a scale-invariant analogue. When κ is observed, the GMM initialization can operate in $\log \kappa$ space, and an optional κ-prior term in the E-step anchors assignments to the known parameter structure.

M-step. With responsibilities fixed, the M-step gives a closed-form update for the mixture weights,

$$\pi_m^{(q)} \leftarrow \frac{N_m^{(q)}}{N}, \qquad N_m^{(q)} = \sum_{i=1}^{N} \gamma_{im}^{(q)}, \tag{16}$$

and a weighted reconstruction problem for each basis:

$$\min_{\Phi_m, \Lambda_m} \sum_{i=1}^{N} \gamma_{im}^{(q)} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2, \tag{17}$$

solved by weighted SVD for TEMPO(POD) or WEIGHTEDNEURALPOD (Algorithm 1) for TEMPO(NeuralPOD). Since gradient descent solves the NeuralPOD subproblem without guaranteeing global optimality, TEMPO is a Generalized EM algorithm [11], which guarantees that the observed-data log-likelihood is non-decreasing at every iteration.

Adaptive basis update. Retraining all M NeuralPOD bases at every EM iteration is computationally prohibitive. We introduce a distribution-shift criterion to determine whether each basis requires updating. Let $\alpha^{(i)}$ denote the global POD projection of $s^{(i)}$, computed once during initialization and fixed thereafter. The responsibility-weighted first and second moments of regime m are

$$\mu_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^{N} \gamma_{im}^{(q)} \, \alpha^{(i)}, \tag{18}$$

$$\Sigma_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^{N} \gamma_{im}^{(q)} \, (\alpha^{(i)} - \mu_m^{(q)})(\alpha^{(i)} - \mu_m^{(q)})^\top. \tag{19}$$

The distribution-shift criterion measures the relative change in these moments between consecutive iterations:

$$\Delta_m^{(q)} = \frac{\|\mu_m^{(q)} - \mu_m^{(q-1)}\|_2}{\|\mu_m^{(q-1)}\|_2 + \epsilon} + \frac{\|\Sigma_m^{(q)} - \Sigma_m^{(q-1)}\|_F}{\|\Sigma_m^{(q-1)}\|_F + \epsilon}. \tag{20}$$

where $\epsilon > 0$ is a small numerical stabiliser. $\Delta_m^{(q)}$ measures how much the composition of regime m has changed, not just the volume of reassignments. Two trajectories with very different $\alpha^{(i)}$ swapping regimes produce a large $\Delta_m^{(q)}$ and trigger a basis update; two similar snapshots doing so do not. Given thresholds $0 < \varepsilon_s < \varepsilon_l$, regime m is SKIPped if $\Delta_m^{(q)} < \varepsilon_s$, updated INCREMENTALly (add or prune one mode) if $\varepsilon_s \leq \Delta_m^{(q)} < \varepsilon_l$, and fully retrained from scratch (FULL RERUN) otherwise. In the INCREMENTAL case, we compute the residual of the current basis under the updated responsibilities $\gamma^{(q)}$, add a new mode if $\sum_i \gamma_{im}^{(q)} \|r_m^{(K_m, i)}\|_w^2 \geq \tau \cdot \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$, and prune the last mode if its weighted contribution falls below ε_p. The FULL RERUN case uses a cold-start random initialization, as warm-starting may trap gradient descent in the geometry of the previous regime.

Convergence and model selection. EM terminates when $\max_m \Delta_m^{(q)} < \varepsilon_c$. The number of regimes $M \in \{2, 3, 4\}$ is selected by BIC on the GMM fit, with mean assignment entropy as a secondary criterion; see Table 3 (Appendix A.1). The complete offline procedure is given in Algorithm 2.

3.4 TEMPO: Online Gated Operator

After Algorithm 2 converges, all regime bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ are frozen. The online phase trains a GatingNet $G : \mathbb{R}^{m_s + d_\kappa} \to \Delta^{M-1}$ that maps (u_0, κ) to regime weights, and

M BranchNets $\mathcal{B}_m : \mathbb{R}^{m_s+d_\kappa} \to \mathbb{R}^{K_m}$ that predict expansion coefficients for each regime. The TEMPO prediction at query point $y = (x,t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(y;\, u_0, \kappa) = \sum_{m=1}^{M} g_m(u_0,\kappa) \left[\phi_0^{(m)}(y) + \sum_{k=1}^{K_m} b_{m,k}(u_0,\kappa)\, \phi_k^{(m)}(y) \right], \tag{21}$$

where $\mathbf{g} = G(u_0,\kappa)$ and $\mathbf{b}_m = \mathcal{B}_m(u_0,\kappa)$. The BranchNet outputs $\mathbf{b}_m$ predict the expansion coefficients $\boldsymbol{\lambda}_m^{(i)}$ for new inputs (u_0,κ) not seen in training. The prediction is mesh-free and evaluable at any $y \in \Omega \times \mathcal{T}$ without retraining.

Training objective. Writing $g_m^{(i)} = g_m(u_0^{(i)}, \kappa^{(i)})$, the online loss is

$$\mathcal{L}_{\text{online}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|s^{(i)} - \hat{s}(\cdot;\, u_0^{(i)}, \kappa^{(i)})\|_w^2}_{\mathcal{L}_{\text{data}}} + \lambda_{\text{KL}} \underbrace{\frac{1}{N}\sum_{i,m} g_m^{(i)} \log \frac{g_m^{(i)}}{\gamma_{im}^*}}_{\mathcal{L}_{\text{KL}}} - \lambda_H \underbrace{\frac{1}{N}\sum_{i,m} g_m^{(i)} \log g_m^{(i)}}_{-\mathcal{L}_{\text{ent}}}. \tag{22}$$

$\mathcal{L}_{\text{data}}$ drives reconstruction accuracy. $\mathcal{L}_{\text{KL}}$ anchors the online gating to the offline regime partition: it penalizes deviation of $\mathbf{g}^{(i)}$ from the EM responsibilities $\boldsymbol{\gamma}_i^*$, so the network cannot discard the discovered structure in pursuit of a simpler routing. $\mathcal{L}_{\text{ent}}$ prevents collapse to a single expert, ensuring all M regime bases contribute. The hyperparameters $\lambda_{\text{KL}}, \lambda_H \geq 0$ trade off these objectives.

4 Computational Experiments

We evaluate TEMPO on parametric PDE benchmarks, verifying three claims: (i) the offline EM phase recovers interpretable, physics-consistent regime structure from unlabelled snapshots; (ii) locally adapted bases achieve lower reconstruction error than a single global basis at equal mode count; (iii) a jointly-trained gated operator with regime-specific bases outperforms a single jointly-trained basis without regime discovery.

4.1 Experimental Setup

Datasets. We use two parametric PDE benchmarks from PDEBench [12].

1D Burgers' equation.

$$u_t + u\, u_x = \nu\, u_{xx}, \qquad x \in (-1,1), \quad t \in (0,2], \tag{23}$$

with periodic boundary conditions and random initial conditions. Viscosity $\nu \in \{0.001,\, 0.01,\, 0.1,\, 1.0\}$ spans four qualitatively distinct solution regimes: smooth diffusive profiles at ν=1.0 and near-discontinuous shock structures at ν=0.001. Each value provides N=9,500 trajectories on N_x=1,024 spatial points and N_t=201 time steps (38,000 total).

2D Darcy flow.

$$-\nabla \cdot \big(a(x)\, \nabla u(x)\big) = \beta, \qquad x \in (0,1)^2, \tag{24}$$

with zero Dirichlet boundary conditions. Here $a(x)$ is a spatially varying permeability field sampled from a random sum of Gaussians, and $\beta \in \{0.01,\, 0.1,\, 1.0,\, 10.0,\, 100.0\}$ is the constant source term. Each value provides N=8,000 training and 1,000 test samples discretised on a 128×128 uniform grid (N_y=16,384 spatial points), 40,000 trajectories in total. The operator maps $a(x)$ to the pressure solution $u(x)$.

Methods. All methods are implemented and trained by us. POD-DeepONet [3] and NeuralPOD-DeepONet (*ours*) are trained one model per ν (or β) value and evaluated in-distribution; they serve as a reference ceiling for jointly-trained approaches. POD-DeepONet trained jointly on all parameter values without regime discovery is the primary baseline for TEMPO. TEMPO(POD) uses responsibility-weighted SVD per regime; TEMPO(NeuralPOD) uses Fourier NeuralPOD bases. Both TEMPO variants train a single model on all parameter values jointly, taking (u_0,κ) as input at inference.

Metric. All methods are evaluated by the mean relative L^2 error ε defined in (6), computed per parameter value and reported in Tables 1–2.

4.2 Main Results

1D Burgers' equation. Table 1 reports the full cross-viscosity error matrix for the Burgers' benchmark. Each per-ν specialist achieves low error in-distribution (diagonal entries), but degrades sharply outside its training regime: for example, a model trained on ν=0.1 reaches 0.048 in-distribution yet 0.461 on ν=0.001, a ten-fold increase. This cross-regime failure motivates joint training with automatic regime discovery. TEMPO is compared against a jointly-trained POD-DeepONet without regime discovery; TEMPO improves on this baseline by learning regime-specific bases.

2D Darcy flow. Table 2 shows the cross-β error matrix for Darcy flow. Per-β specialists fail even more severely than in Burgers: the Darcy solution scales linearly with β, so a model trained on one value predicts completely wrong amplitudes on others. Errors grow by roughly one order of magnitude per decade in β, reaching >1000 for the most distant pair. TEMPO trains jointly on $\beta \in \{0.1, 1, 10, 100\}$ and recovers reasonable accuracy across all regimes.

Table 1: **Relative L^2 test error on 1D Burgers' equation.** Each per-ν specialist is trained on a single viscosity and evaluated on all four; underlined entries are in-distribution. The large off-diagonal errors motivate joint training with regime discovery. TEMPO trains jointly on $\nu \in \{0.001, 0.1, 1.0\}$; the ν=0.01 column is omitted for jointly-trained methods (—). **Bold**: best per column among jointly-trained methods.

Method	Train ν	ν=0.001	ν=0.01	ν=0.1	ν=1.0
Per-ν specialists (single-viscosity training)					
POD-DeepONet [3]	0.001	<u>0.273</u>	0.215	0.297	0.639
	0.01	0.274	<u>0.214</u>	0.274	0.614
	0.1	0.349	0.299	<u>0.048</u>	0.344
	1.0	0.461	0.425	0.260	<u>0.025</u>
NeuralPOD-DeepONet *(ours)*	0.001	<u>0.392</u>	0.352	0.288	0.535
	0.01	0.391	<u>0.351</u>	0.284	0.541
	0.1	0.426	0.388	<u>0.229</u>	0.371
	1.0	0.483	0.451	0.315	<u>0.183</u>
Joint training ($\nu \in \{0.001, 0.1, 1.0\}$)					
POD-DeepONet [3]	all	—	—	—	—
TEMPO(POD) *(ours)*	M=3	**0.317**	—	**0.168**	**0.182**
TEMPO(NeuralPOD) *(ours)*	M=3	0.571	—	0.502	0.376

4.3 Regime Discovery

Number of regimes. We select M=3 based on BIC and an entropy elbow: the mean assignment entropy H gains $+0.30$ from M=2 to M=3 but only $+0.18$ from M=3 to M=4, indicating the third regime captures genuine structure that a fourth does not (Table 3, Appendix A.1). The UMAP embedding (Figure 2) confirms three distinct manifold branches, one per discovered regime.

Gating structure. The per-viscosity EM responsibilities closely match the global mixture weights $\boldsymbol{\pi}$=(0.324, 0.267, 0.409) across all ν (Table 4, Appendix A.1), confirming that regime membership is driven by the geometry of u_0, not by ν.

Regime structure. Figure 2 shows that each manifold branch spans multiple ν values, confirming the partition reflects solution geometry rather than the physical parameter directly.

Table 2: **Relative L^2 test error on 2D Darcy flow.** Each per-β specialist is trained on a single source value and evaluated on all five; underlined entries are in-distribution. Off-diagonal errors exceed 1 because the Darcy solution scales with β, so a specialist trained on β=100 predicts amplitudes $\sim 10^4$ times larger than the ground truth at β=0.01. TEMPO trains jointly on $\beta \in \{0.1, 1, 10, 100\}$; the β=0.01 regime is excluded from joint training (—). **Bold**: best per column among jointly-trained methods.

Method	Train β	β=0.01	β=0.1	β=1.0	β=10	β=100
Per-β specialists (single-value training)						
	0.01	0.449	0.791	0.961	0.996	>1
	0.1	>1	0.148	0.871	0.986	0.999
POD-DeepONet [3]	1.0	>10	>1	0.048	0.897	0.990
	10	>100	>10	>1	0.035	0.900
	100	>1000	>100	>10	>1	0.042
Joint training ($\beta \in \{0.1, 1, 10, 100\}$)						
POD-DeepONet [3]	all	—	2.557	0.363	0.181	0.347
TEMPO(POD) *(ours)*	M=5	—	**0.775**	**0.178**	**0.136**	**0.073**
TEMPO(NeuralPOD) *(ours)*	M=5	—	3.813	0.549	0.322	0.311

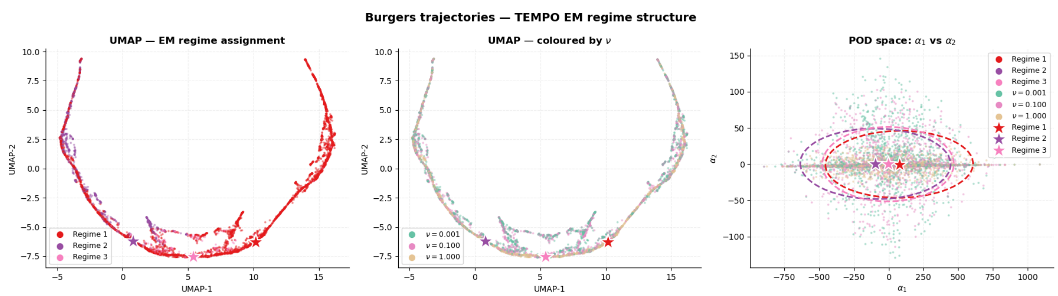

Figure 2: TEMPO regime structure on 1D Burgers' (M=3, 19 EM iterations). *Left:* UMAP coloured by regime. *Centre:* same embedding coloured by ν. *Right:* GMM fit in POD coefficient space (α_1, α_2); stars mark regime centroids μ_m.

5 Conclusion

Operator learning typically uses a single global representation across the full parameter space. When the governing parameter drives qualitative change in solution structure, a single basis must fit incompatible geometries simultaneously, concentrating error where accurate prediction matters most.

TEMPO addresses this by treating regime membership as a latent variable. EM discovers M regimes from unlabelled snapshots and fits a dedicated basis to each, without regime supervision. On 1D Burgers' across three orders of viscosity, the discovered regimes cross viscosity boundaries, confirming that the partition reflects solution geometry rather than the physical parameter directly. Regime-specific bases reduce reconstruction error relative to a global basis at equal mode count. The gated online operator outperforms a single joint baseline across the full parameter range.

Several directions remain open. Conditioning the basis functions on κ would allow each mode to encode parameter-dependent structure directly. Extending TEMPO to higher-dimensional benchmarks will test whether regime discovery generalises beyond one-dimensional shock dynamics. Convergence guarantees for Generalized EM under nonlinear basis parameterisation remain an open theoretical problem.

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A Appendix

A.1 Additional Experiments

Regime model selection (1D Burgers'). Table 3 reports BIC, mean assignment entropy H, and mean test error $\bar{\varepsilon}$ for $M \in \{2, 3, 4\}$. Table 4 shows the per-viscosity EM responsibilities alongside global mixture weights π_m; near-uniform agreement across ν confirms geometry-driven rather than parameter-driven regime membership.

Table 3: Regime count selection on 1D Burgers'. BIC: EM log-likelihood (higher is better); H: mean assignment entropy; $\bar{\varepsilon}$: mean relative L^2 test error.

M	BIC ($\times 10^4$)	H	$\bar{\varepsilon}$
2	3.90	0.622	0.209
3	**4.92**	**0.920**	**0.222**
4	5.47	1.099	0.228

Table 4: Mean EM assignment weights per viscosity (1D Burgers', TEMPO(POD), M=3) vs. global mixture weights π_m. Each row sums to 1; near-uniform agreement across ν confirms regime membership is driven by u_0 geometry, not by ν.

ν	Regime 1	Regime 2	Regime 3
0.001	0.32	0.28	0.40
0.1	0.32	0.26	0.42
1.0	0.33	0.26	0.41
π_m	0.32	0.27	0.41

POD basis quality (1D Burgers'). Figure 3 shows the singular value spectrum and cumulative explained variance for the global POD on 1D Burgers'. K=30 modes capture 99.99% of the total variance, achieving a Phase 1 relative L^2 reconstruction error of 1.47%.

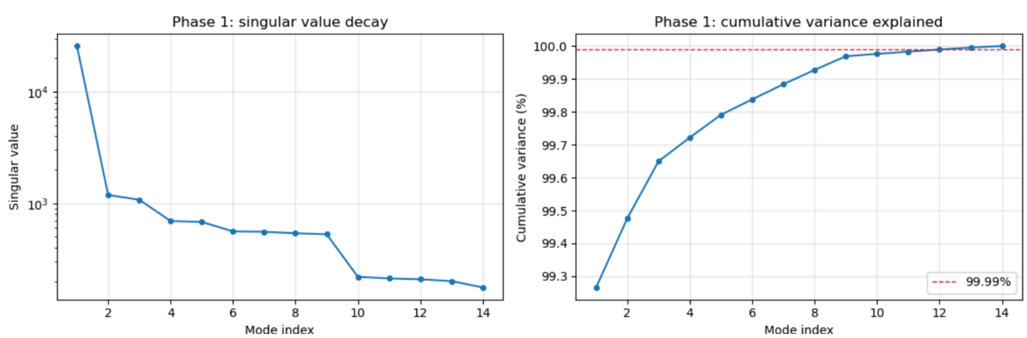

Figure 3: Global POD basis on 1D Burgers' (ν=1.0): singular value spectrum (*left*) and cumulative explained variance (*right*); K=30 modes attain 99.99%.

NeuralPOD basis quality (1D Burgers'). Figure 4 shows the greedy residual norm as a function of mode count; the steep initial decay confirms that the Fourier NeuralPOD basis concentrates variance efficiently. Figure 5 shows the learned spatial modes alongside representative reconstructions; modes are smooth and globally supported for large ν, and sharply localised near the shock front for small ν.

POD vs. NeuralPOD basis comparison (1D Burgers', ν=1.0). Figure 6 shows the singular-value decay of the global POD basis alongside the weighted residual decay of the Fourier Neural-POD basis. Figure 7 shows the first K spatiotemporal mode functions for both bases; POD modes are globally supported, while NeuralPOD modes adapt their spatial localisation to the data.

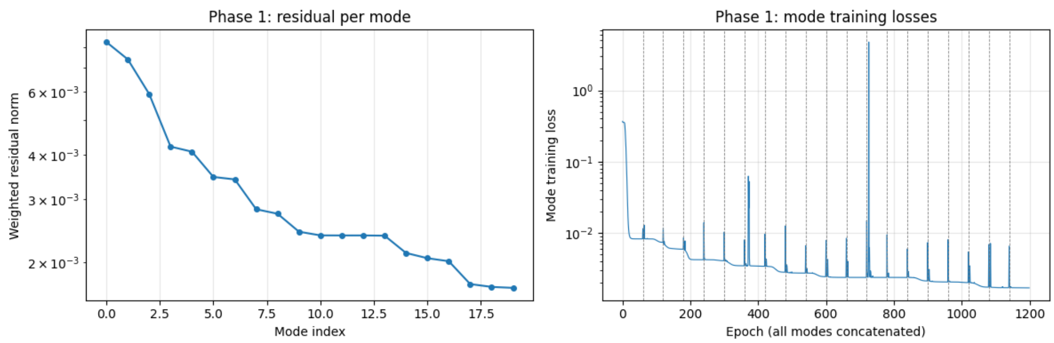

Figure 4: Fourier NeuralPOD basis on 1D Burgers' (ν=1.0): weighted residual norm versus greedy mode count.

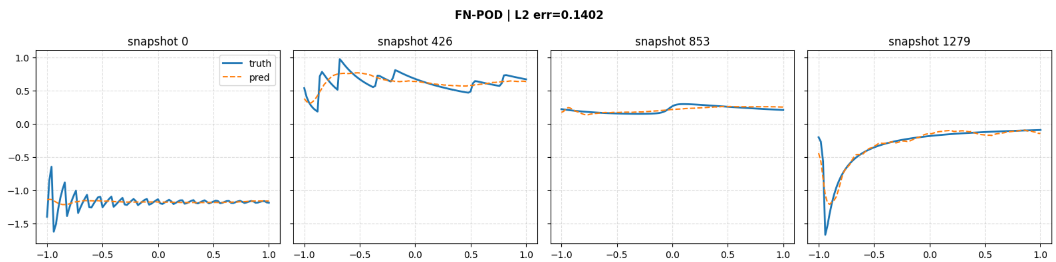

Figure 5: Learned Fourier NeuralPOD basis functions and representative reconstructions on 1D Burgers' (ν=1.0).

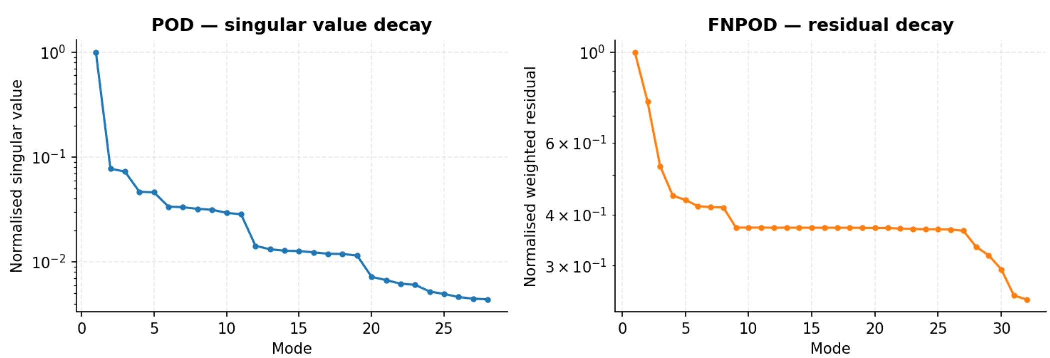

Figure 6: POD singular-value decay (*left*) and Fourier NeuralPOD residual decay (*right*) on 1D Burgers' (ν=1.0).

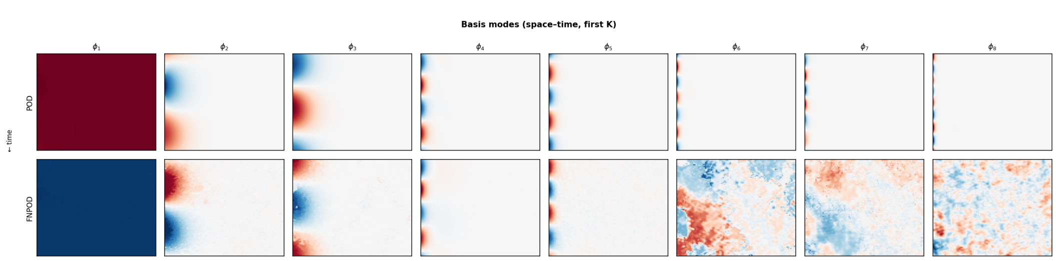

Figure 7: First K basis modes on 1D Burgers' (ν=1.0): POD (*top*) vs. Fourier NeuralPOD (*bottom*).

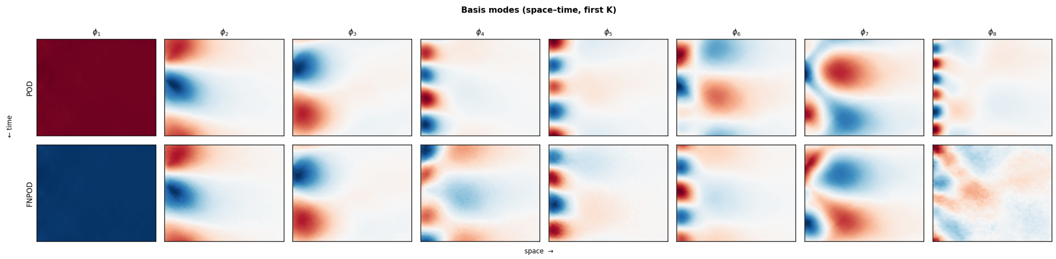

Figure 8: First K basis modes on 1D Burgers' (ν=0.01): POD (*top*) vs. Fourier NeuralPOD (*bottom*).

Baseline training curves. Figures 9 and 10 show the Phase 2 MSE training curves for POD-DeepONet and NeuralPOD-DeepONet on 1D Burgers' (ν=1.0).

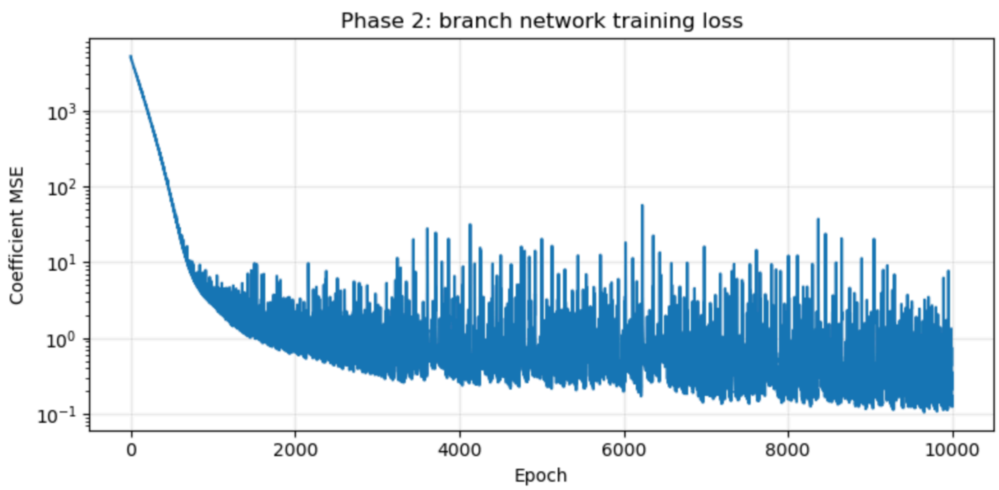

Figure 9: Phase 2 MSE training curve for POD-DeepONet on 1D Burgers' (ν=1.0).

Figure 10: Phase 2 MSE training curve for NeuralPOD-DeepONet on 1D Burgers' (ν=1.0).

Error distributions. Figures 11 and 12 show the full test-set error distributions for POD-DeepONet and NeuralPOD-DeepONet on 1D Burgers' (ν=1.0).

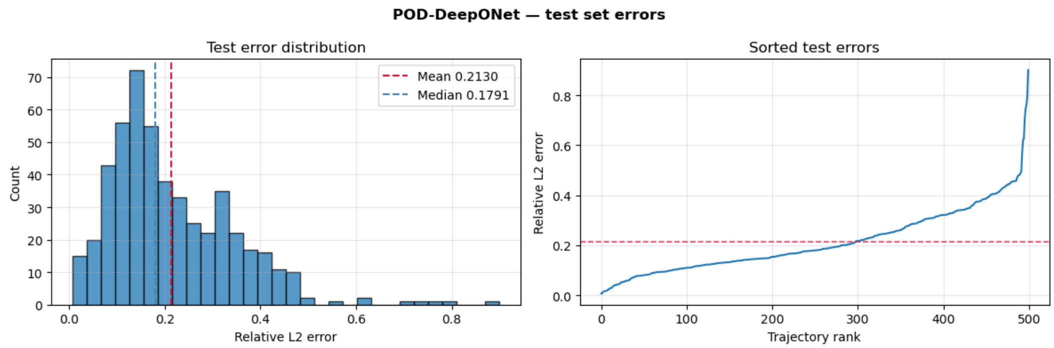

Figure 11: POD-DeepONet test-set error distribution on 1D Burgers' (ν=1.0): error histogram (*left*) and sorted errors (*right*).

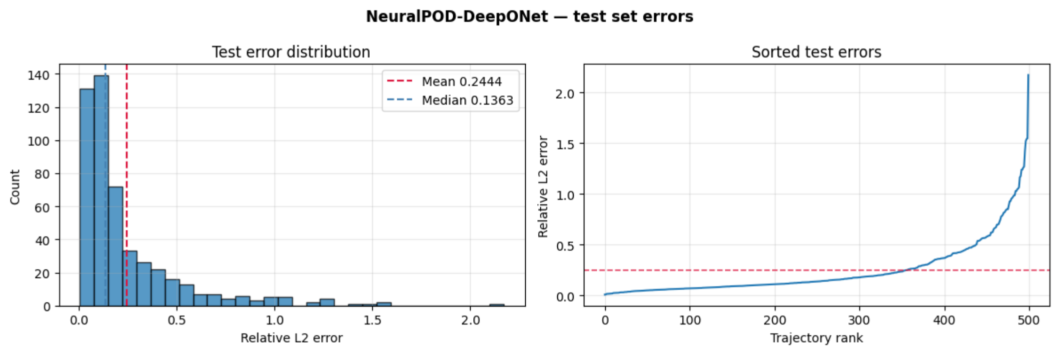

Figure 12: NeuralPOD-DeepONet test-set error distribution on 1D Burgers' (ν=1.0). Layout as in Figure 11.

Representative reconstructions. Figures 13 and 14 show representative space–time predictions for POD-DeepONet and NeuralPOD-DeepONet on 1D Burgers' (ν=1.0).

TEMPO reconstructions. Figures 15–18 show representative predictions from all four TEMPO variants.

TEMPO regime structure (2D Darcy flow, M=5). Figure 19 shows the TEMPO Phase 1 regime structure on 2D Darcy flow with M=5 regimes.

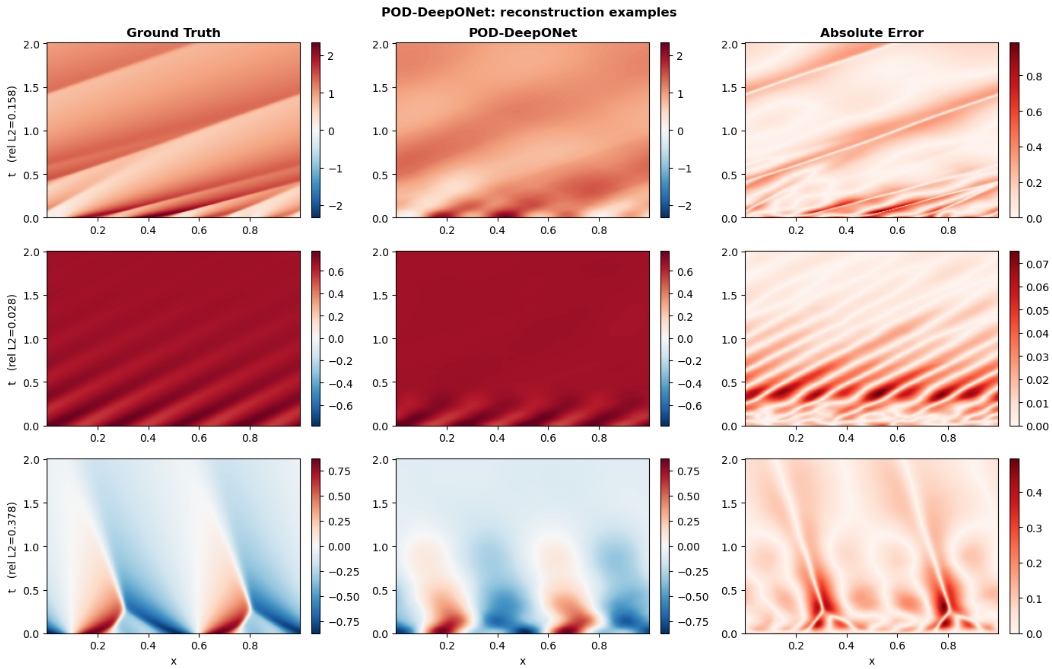

Figure 13: POD-DeepONet: representative reconstructions on 1D Burgers' (ν=1.0). Each row: ground truth (*left*), prediction (*centre*), point-wise absolute error (*right*).

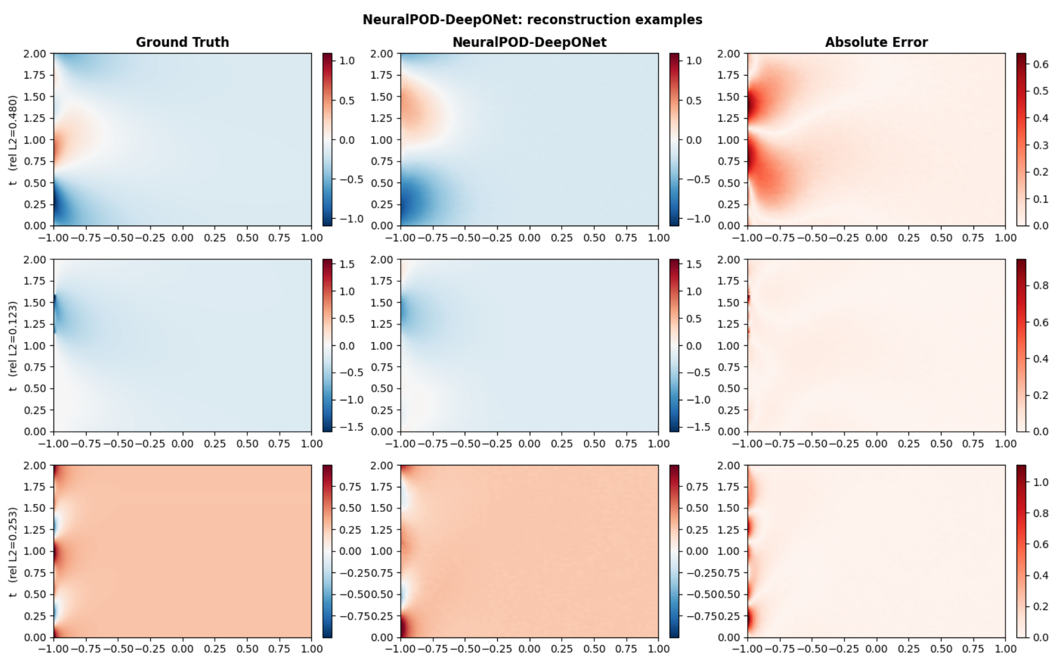

Figure 14: NeuralPOD-DeepONet: representative reconstructions on 1D Burgers' (ν=1.0). Layout as in Figure 13.

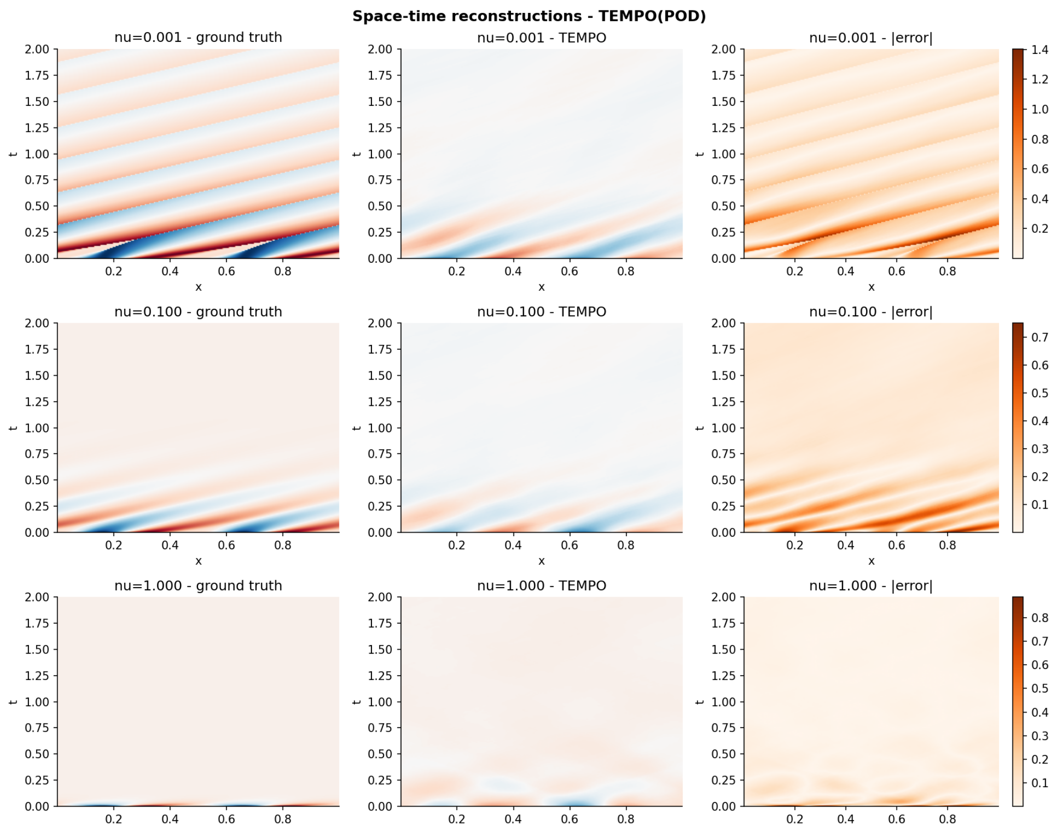

Figure 15: TEMPO(POD) on 1D Burgers' (M=3): ground truth (*left*), prediction (*centre*), pointwise absolute error (*right*).

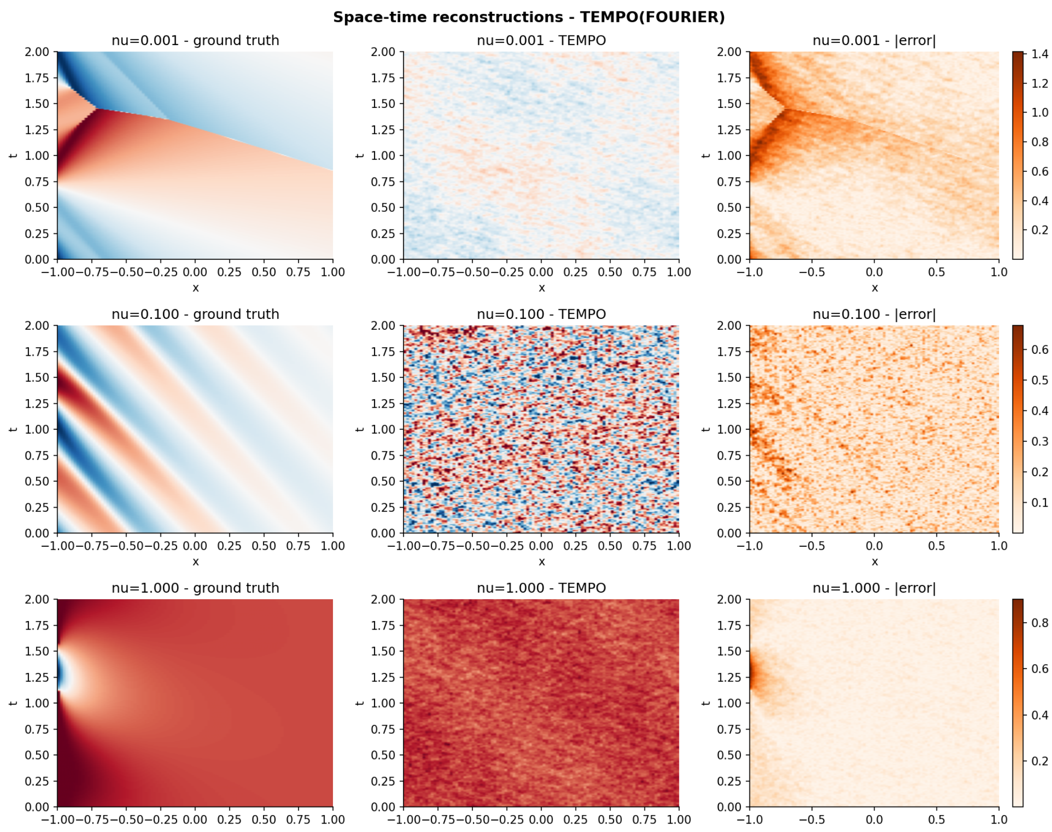

Figure 16: TEMPO(NeuralPOD) on 1D Burgers' (M=3). Layout as in Figure 15.

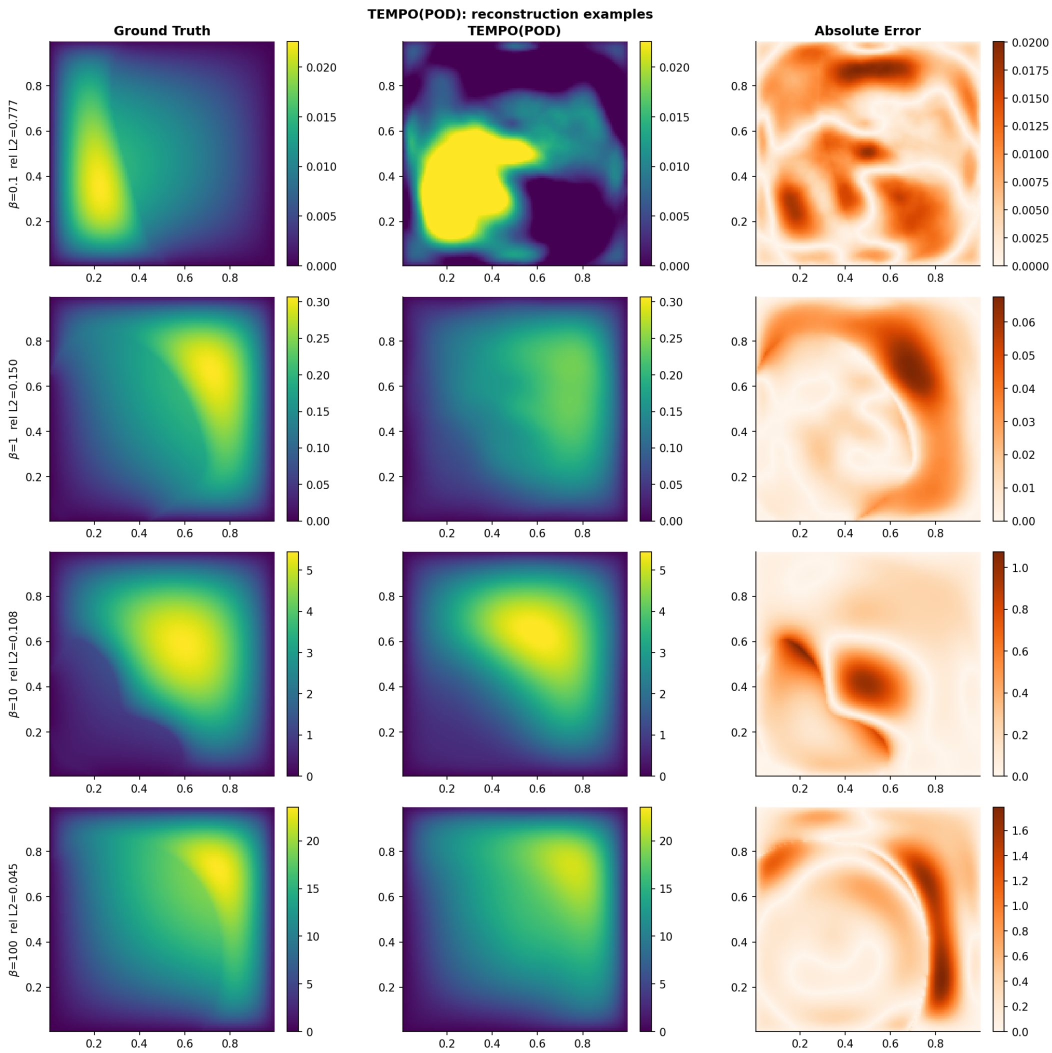

Figure 17: TEMPO(POD) on 2D Darcy flow ($M=5$): ground truth (*left*), prediction (*centre*), point-wise absolute error (*right*).

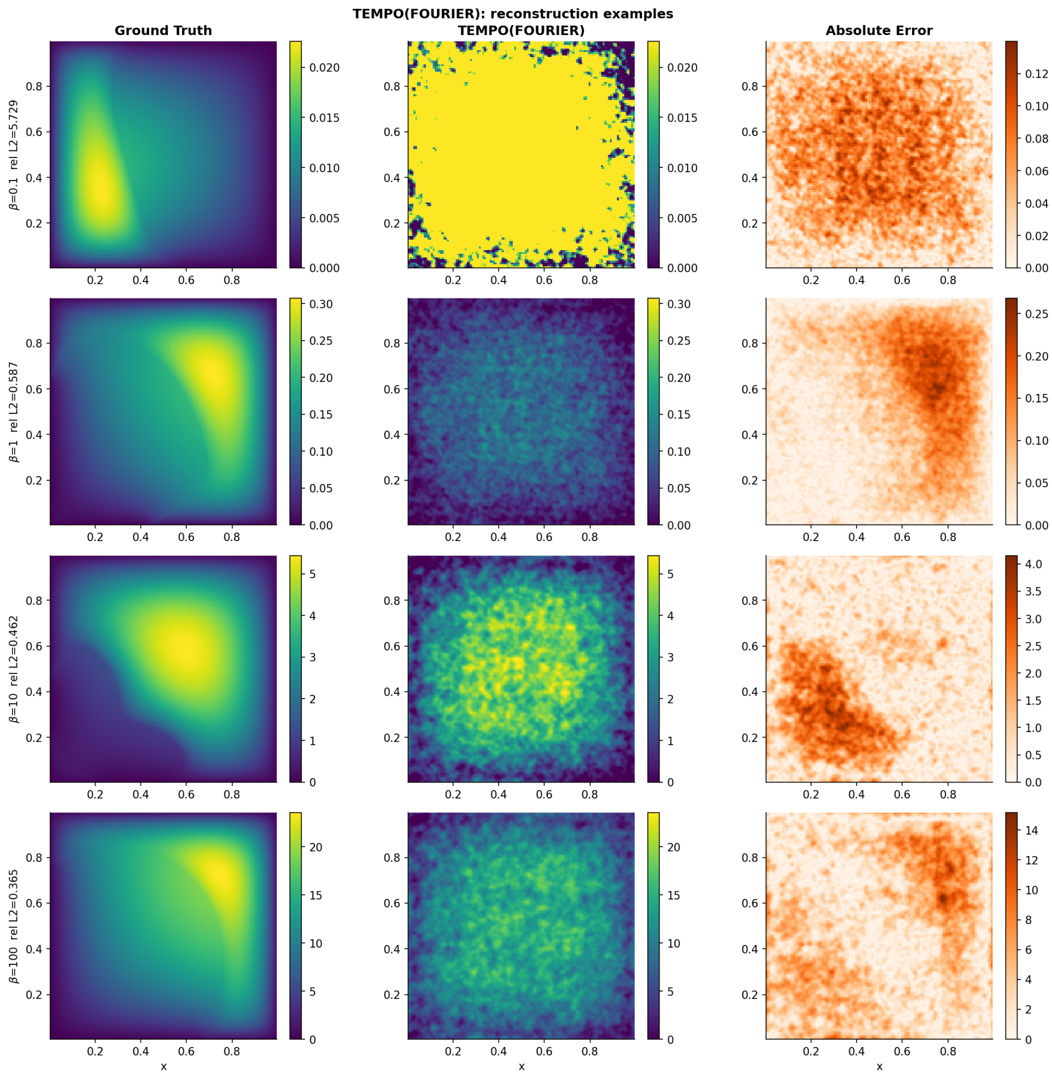

Figure 18: TEMPO(NeuralPOD) on 2D Darcy flow (M=5). Layout as in Figure 17.

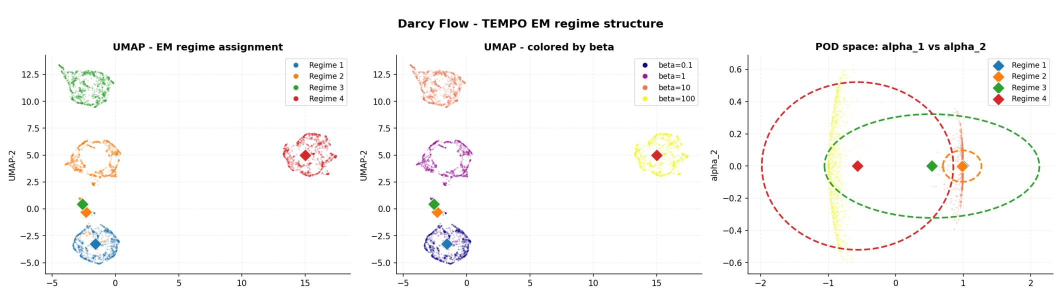

Figure 19: TEMPO regime structure on 2D Darcy flow (M=5): UMAP coloured by regime assignment (*left*) and by β (*right*).

A.2 Algorithms

Algorithm 1 $\textsc{WeightedNeuralPOD}\big(\{s^{(i)}, \gamma_{im}\}_{i=1}^{N},\ \tau\big)$

$\bar{s}^{(m)} \leftarrow \sum_{i=1}^{N} \gamma_{im}\, s^{(i)}$ ▷ responsibility-weighted mean trajectory

Train $\phi_0^{(m)}$ by minimizing $\sum_j w_j \big(\phi_0^{(m)}(y_j) - \bar{s}_j^{(m)}\big)^2$

$r^{(0,i)} \leftarrow s^{(i)} - \phi_0^{(m)}(y)$ for all i; $p \leftarrow 0$

$V_m \leftarrow \sum_{i=1}^{N} \gamma_{im}\, \|s^{(i)} - \bar{s}^{(m)}\|_w^2$ ▷ initial unmodeled variance

while $\sum_{i=1}^{N} \gamma_{im}\, \|r^{(p,i)}\|_w^2 \ \geq\ \tau\, V_m$ **and** $p < P_{\max}$ **do**

  Jointly minimize over $\phi_{p+1}^{(m)}$ and $\{\lambda_{p+1}^{(m,i)}\}_{i=1}^{N}$:

$$\sum_{i=1}^{N} \gamma_{im} \sum_j w_j \Big(\lambda_{p+1}^{(m,i)}\, \phi_{p+1}^{(m)}(y_j) - r_j^{(p,i)}\Big)^2$$

  $r^{(p+1,i)} \leftarrow r^{(p,i)} - \lambda_{p+1}^{(m,i)}\, \phi_{p+1}^{(m)}(y)$ for all i

  $p \leftarrow p + 1$

end while

return $P_m \leftarrow p,\ \phi_0^{(m)},\ \big\{\phi_p^{(m)},\ \{\lambda_p^{(m,i)}\}_{i=1}^{N}\big\}_{p=1}^{P_m}$

Algorithm 2 TEMPO — Offline Phase

Require: $\mathcal{D} = \{u_0^{(i)}, \kappa^{(i)}, s^{(i)}\}_{i=1}^{N};\ M,\ \varepsilon_s,\ \varepsilon_l,\ \varepsilon_p,\ \varepsilon_c,\ \tau$
Ensure: Regime bases $\{\Phi_m^*\}$, amplitudes $\{\Lambda_m^*\}$, responsibilities $\{\gamma_{im}^*\}$, weights $\{\pi_m^*\}$

Initialization
Compute global POD on $\{s^{(i)}\}$; project to $\alpha^{(i)} \in \mathbb{R}^P$ for all i
Run GMM-EM on $\{\alpha^{(i)}\} \to \gamma_{im}^{(0)},\ \pi_m^{(0)},\ \mu_m^{(0)},\ \Sigma_m^{(0)}$
for $m = 1, \ldots, M$ **do**
    $\Phi_m^{(0)}, \Lambda_m^{(0)} \leftarrow$ WEIGHTEDNEURALPOD($\{s^{(i)}, \gamma_{im}^{(0)}\}, \tau$)   ▷ or weighted SVD for TEMPO(POD), see §3.1
end for
$\sigma^2 \leftarrow (2N)^{-1} \sum_{i,m} \gamma_{im}^{(0)} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2$ (14)
EM iterations
repeat
    E-step
    for all $i = 1, \ldots, N$ and $m = 1, \ldots, M$ **do**
        Compute $\hat{s}_m^{(i)}$ via (**??**) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$
        $\gamma_{im}^{(q)} \leftarrow$ Eq. (15)
    end for
    M-step: mixture weights
    $\pi_m^{(q)} \leftarrow N_m^{(q)}/N$ for all m (16)
    M-step: bases
    for $m = 1, \ldots, M$ **do**
        Compute $\mu_m^{(q)}$ (18), $\Sigma_m^{(q)}$ (19), $\Delta_m^{(q)}$ (20)
        if $\Delta_m^{(q)} < \varepsilon_s$ **then**
            $\Phi_m^{(q)} \leftarrow \Phi_m^{(q-1)}$ SKIP
        else if $\Delta_m^{(q)} < \varepsilon_l$ **then**   INCREMENTAL
            Compute $r_m^{(P_m)}$ under updated $\gamma^{(q)}$; $V_m^{(q)} \leftarrow \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$
            if $\sum_i \gamma_{im}^{(q)} \|r_m^{(P_m,i)}\|_w^2 \geq \tau \cdot V_m^{(q)}$ **then**
                Train mode $P_m{+}1$; $P_m \leftarrow P_m + 1$
            end if
            if $\|\phi_{P_m}^{(m)}\|_w^2 \cdot \sum_{i=1}^{N} \gamma_{im}^{(q)} \left(\lambda_{P_m}^{(m,i)}\right)^2 < \varepsilon_p$ **then**
                $P_m \leftarrow P_m - 1$
            end if
        else   FULL RERUN
            $\Phi_m^{(q)}, \Lambda_m^{(q)} \leftarrow$ WEIGHTEDNEURALPOD($\{s^{(i)}, \gamma_{im}^{(q)}\}, \tau$)
        end if
    end for
until $\max_m \Delta_m^{(q)} < \varepsilon_c$
return $\Phi_m^* \leftarrow \Phi_m^{(q)},\ \Lambda_m^* \leftarrow \Lambda_m^{(q)},\ \gamma_{im}^* \leftarrow \gamma_{im}^{(q)},\ \pi_m^* \leftarrow \pi_m^{(q)}$